

Formula generală pentru produsul tensorial a doi vectori este

$$\mathbf{a} \otimes \mathbf{b} = \mathbf{b} \cdot \mathbf{a}^t.$$

Astfel, în plan

$$\mathbf{a} \otimes \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \cdot \begin{pmatrix} a_1 & a_2 \end{pmatrix} = \begin{pmatrix} a_1 b_1 & a_2 b_1 \\ a_1 b_2 & a_2 b_2 \end{pmatrix},$$

iar în spațiu

$$\mathbf{a} \otimes \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \cdot \begin{pmatrix} a_1 & a_2 & a_3 \end{pmatrix} = \begin{pmatrix} a_1 b_1 & a_2 b_1 & a_3 b_1 \\ a_1 b_2 & a_2 b_2 & a_3 b_2 \\ a_1 b_3 & a_2 b_3 & a_3 b_3 \end{pmatrix}.$$

Transformări plane

Translația în plan

$$\text{Trans}(\mathbf{w}) = \begin{bmatrix} 1 & 0 & w_1 \\ 0 & 1 & w_2 \\ 0 & 0 & 1 \end{bmatrix}. \quad (1)$$

Rotația

$$\text{Rot}(Q, \theta) = \begin{bmatrix} \cos \theta & -\sin \theta & q_1(1 - \cos \theta) + q_2 \sin \theta \\ \sin \theta & \cos \theta & -q_1 \sin \theta + q_2(1 - \cos \theta) \\ 0 & 0 & 1 \end{bmatrix}. \quad (2)$$

Scalarea uniformă

$$\text{Scale}(Q, s) = \begin{pmatrix} s & 0 & (1-s)q_1 \\ 0 & s & (1-s)q_2 \\ 0 & 0 & 1 \end{pmatrix}. \quad (3)$$

Scalarea neuniformă

$$\text{Scale}(Q, s_x, s_y) = \begin{pmatrix} s_x & 0 & (1-s_x)q_1 \\ 0 & s_y & (1-s_y)q_2 \\ 0 & 0 & 1 \end{pmatrix}. \quad (4)$$

Reflexia

$$\text{Mirror}(Q, \mathbf{w}) = \begin{pmatrix} I_2 - 2(\mathbf{w}^\perp \otimes \mathbf{w}^\perp) & 2(\mathbf{w}^\perp \otimes \mathbf{w}^\perp) \cdot Q \\ 0 & 1 \end{pmatrix}. \quad (5)$$

Forfecarea

$$\text{Shear}(Q, \mathbf{w}, \theta) = \begin{pmatrix} I_2 + \text{tg } \theta (\mathbf{w}^\perp \otimes \mathbf{w}) & -\text{tg } \theta (\mathbf{w}^\perp \otimes \mathbf{w}) \cdot Q \\ 0 & 1 \end{pmatrix}. \quad (6)$$

Transformări în spațiu

Translația

$$\text{Trans}(\mathbf{w}) = \begin{pmatrix} 1 & 0 & 0 & w_1 \\ 0 & 1 & 0 & w_2 \\ 0 & 0 & 1 & w_3 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (7)$$

Rotația

$$\text{Rot}(Q, \mathbf{u}, \theta) = \begin{pmatrix} \text{Rot}(\mathbf{u}, \theta) & (I_3 - \text{Rot}(\mathbf{u}, \theta)) \cdot Q \\ 0 & 1 \end{pmatrix}, \quad (8)$$

unde

$$\text{Rot}(\mathbf{u}, \theta) = \cos \theta \cdot I_3 + (1 - \cos \theta)(\mathbf{u} \otimes \mathbf{u}) + \sin \theta \cdot (\mathbf{u} \times -) \quad \text{și} \quad \mathbf{u} \times - = \begin{pmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{pmatrix}.$$

Scalarea uniformă

$$\text{Scale}(Q, s) = \begin{pmatrix} s & 0 & 0 & (1-s) \cdot q_x \\ 0 & s & 0 & (1-s) \cdot q_y \\ 0 & 0 & s & (1-s) \cdot q_z \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (9)$$

Scalarea neuniformă

$$\text{Scale}(Q, s_x, s_y, s_z) = \begin{pmatrix} s_x & 0 & 0 & (1-s_x) \cdot q_x \\ 0 & s_y & 0 & (1-s_y) \cdot q_y \\ 0 & 0 & s_z & (1-s_z) \cdot q_z \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (10)$$

Reflexia

$$\text{Mirror}(Q, \mathbf{n}) = \begin{pmatrix} I_3 - 2(\mathbf{n} \otimes \mathbf{n}) & 2(\mathbf{n} \otimes \mathbf{n}) \cdot Q \\ 0 & 1 \end{pmatrix} \quad (11)$$

Forfecarea

$$\text{Shear}(Q, \mathbf{n}, \mathbf{u}, \theta) = \begin{pmatrix} I_3 + \text{tg } \theta \cdot (\mathbf{n} \otimes \mathbf{u}) & -\text{tg } \theta \cdot (\mathbf{n} \otimes \mathbf{u}) \cdot Q \\ 0 & 1 \end{pmatrix}. \quad (12)$$